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Unit: 01 Differential Calculus

TOPICS: 1) successive diff.

2) Leibnitz's thm.

3) Taylor's & Maclaurin's expansion

4) Indeterminate forms, L-Hospital rule

Standard Results

$$1) \frac{d}{dx} x^n = nx^{n-1}$$

$$2) \frac{d}{dx} \log x = \frac{1}{x}$$

$$3) \frac{d}{dx} e^x = e^x$$

$$4) \frac{d}{dx} a^x = a^x \log a$$

$$5) \frac{d}{dx} \sin x = \cos x$$

$$6) \frac{d}{dx} \cos x = -\sin x$$

$$7) \frac{d}{dx} \tan x = \sec^2 x$$

$$8) \frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$$

$$9) \frac{d}{dx} \sec x = \sec x \tan x$$

$$10) \frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x$$

$$11) \frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$$

$$12) \frac{d}{dx} \tan^{-1} x = \frac{1}{x^2+1}$$

$$13) \frac{d}{dx} \sec^{-1} x = \frac{1}{x\sqrt{x^2-1}}$$

$$14) \frac{d}{dx} (uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$15) \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{d}{dx}(u) - u \frac{d}{dx}(v)}{v^2}$$

$$16) \frac{d}{dx} \frac{1}{x^n} = -\frac{n}{x^{n+1}}$$

n^{th} order derivatives of some standard fns:-

$$1) \frac{d^n}{dx^n} (ax+b)^m = D^n (ax+b)^m = m(m-1)(m-2)\dots(m-n+1) \times a^n \times (ax+b)^{m-n}$$

proof:- $D(ax+b)^m = m(ax+b)^{m-1} \cdot a$

$$D^2(ax+b)^m = m(m-1)(ax+b)^{m-2} a^2$$

$$D^3(ax+b)^m = m(m-1)(m-2)(ax+b)^{m-3} a^3$$

$$\therefore D^n(ax+b)^m = m(m-1)(m-2)\dots[m-(n-1)](ax+b)^{m-n} a^n$$

$$2) D^n \frac{1}{(ax+b)^{n+1}} = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

$$3) D^n \log(ax+b) = \frac{(-1)^{n-1} (n-1)! a^n}{(ax+b)^n}$$

$$4) D^n (a^{mx}) = m^n (\log a)^n a^{mx}$$

$$\text{proof: } \Rightarrow D(a^{mx}) = a^{mx} \log a \cdot m$$

$$D^2(a^{mx}) = a^{mx} (\log a)^2 m^2$$

$$D^n(a^{mx}) = a^{mx} (\log a)^n m^n$$

$$5) D^n(e^{ax}) = a^n e^{ax}$$

$$6) D^n \sin(ax+b) = a^n \sin(ax+b + n \frac{\pi}{2})$$

$$7) D^n \cos(ax+b) = a^n \cos(ax+b + n \frac{\pi}{2})$$

$$8) D^n [e^{ax} \sin(bx+c)] = (a^2+b^2)^{n/2} e^{ax} \sin(bx+c + n \tan^{-1} \frac{b}{a})$$

$$9) D^n [e^{ax} \cos(bx+c)] = (a^2+b^2)^{n/2} e^{ax} \cos(bx+c + n \tan^{-1} \frac{b}{a})$$

$$\text{To prove: } D^n [e^{ax} \sin(bx+c)] = (a^2+b^2)^{n/2} e^{ax} \sin(bx+c + n \tan^{-1} \frac{b}{a})$$

$$\Rightarrow D[e^{ax} \sin(bx+c)] = e^{ax} b \cos(bx+c) + e^{ax} a \sin(bx+c)$$

$$= e^{ax} [a \sin(bx+c) + b \cos(bx+c)]$$

$$\text{put } a = r \cos \alpha, \quad b = r \sin \alpha$$

$$\therefore r = \sqrt{a^2+b^2}, \quad \alpha = \tan^{-1} \frac{b}{a}$$

$$y_1 = z e^{ax} [\sin(bx+c) \cos \alpha + \cos(bx+c) \sin \alpha]$$

$$y_1 = z e^{ax} \sin(bx+c+\alpha)$$

$$y_2 = z^2 e^{ax} \sin(bx+c+2\alpha)$$

$$y_3 = z^3 e^{ax} \sin(bx+c+3\alpha)$$

$$y_n = z^n e^{ax} \sin(bx+c+n\alpha)$$

$$y_n = (a^2 + b^2)^{n/2} e^{ax} \sin[bx+c+n \tan^{-1} \frac{b}{a}]$$

e.g. ① If $y = x \log \frac{x-1}{x+1}$ S.T. $y_n = (-1)^{n-2} (n-2)! \times$

$$\left[\frac{x-n}{(x-1)^n} - \frac{x+n}{(x+1)^n} \right]$$

$$\Rightarrow y = x \log \frac{x-1}{x+1}$$

$$y_1 = \log \frac{x-1}{x+1} + x \left[\frac{1}{x-1} - \frac{1}{x+1} \right]$$

$$y_1 = \log(x-1) - \log(x+1) + x \left[\frac{1}{x-1} - \frac{1}{x+1} \right]$$

$$= \log(x-1) - \log(x+1) + \frac{x-1-x}{x-1} + \frac{x+1-x}{x+1}$$

$$= \log(x-1) - \log(x+1) + 1 + \frac{1}{x-1} - 1 + \frac{1}{x+1}$$

$$y_1 = \log(x-1) - \log(x+1) + \frac{1}{x-1} + \frac{1}{x+1}$$

Diff. $(n-1)$ times w.r.t. x

$$y_n = \frac{(-1)^{n-2} (n-2)!}{(x-1)^{n-1}} - \frac{(-1)^{n-2} (n-2)!}{(x+1)^{n-1}} + \frac{(-1)^{n-1} (n-1)!}{(x+1)^n} \cdot 1$$

$$+ \frac{(-1)^{n-1} (n-1)!}{(x+1)^n} \cdot 1$$

$$= (-1)^{n-2} (n-2)! \left[\frac{1}{(x-1)^{n-1}} - \frac{1}{(x+1)^{n-1}} + \frac{(n-1)}{(x+1)^n} - \frac{(n-1)}{(x+1)^n} \right]$$

$$= (-1)^{n-2} (n-2)! \left[\frac{x-1}{(x-1)^n} - \frac{(x+1)}{(x+1)^n} - \frac{(n-1)}{(x-1)^n} + \frac{(n-1)}{(x+1)^n} \right]$$

$$y_n = (-1)^{n-2} (n-2)! \left[\frac{x-n}{(x-1)^n} - \frac{x+n}{(x+1)^n} \right]$$

② Find n^{th} order derivatives of

i) $\cos x \cos 2x \cos 3x$

ii) $e^{2x} \cos^2 x \sin x$

i) $y = \cos x \cos 2x \cos 3x$

$$y = \frac{1}{2} \cos x [\cos 5x + \cos x]$$

$$= \frac{1}{4} [\cos 6x + \cos 4x + \cos 2x + 1]$$

$$y = \frac{1}{4} [1 + \cos 2x + \cos 4x + \cos 6x]$$

$$y_n = \frac{1}{4} \left[2^n \cos(2x + n\frac{\pi}{2}) + 4^n \cos(4x + n\frac{\pi}{2}) + 6^n \cos(6x + n\frac{\pi}{2}) \right]$$

$$11) y = e^{2x} \cos^2 x \sin x$$

$$y = e^{2x} \sin x \cdot \frac{1}{2} (1 + \cos 2x)$$

$$y = \frac{1}{2} [e^{2x} \sin x + e^{2x} \sin x \cos 2x]$$

$$y = \frac{1}{2} [e^{2x} \sin x + \frac{1}{2} e^{2x} (\sin 3x - \sin x)]$$

$$= \frac{1}{2} e^{2x} \sin x + \frac{1}{4} e^{2x} \sin 3x - \frac{1}{4} e^{2x} \sin x$$

$$y = \frac{1}{4} e^{2x} \sin 3x + \frac{1}{4} e^{2x} \sin x$$

$$y_n = \frac{1}{4} (2^2 + 3^2)^{n/2} \sin(3x + n \tan^{-1} \frac{3}{2}) + \frac{1}{4} (2^2 + 1^2)^{n/2} \sin(x + n \tan^{-1} \frac{1}{2})$$

$$y_n = \frac{13^{n/2}}{4} \sin(3x + n \tan^{-1} \frac{3}{2}) + \frac{5^{n/2}}{4} \sin(x + n \tan^{-1} \frac{1}{2})$$

⑧ Find n^{th} order derivative of

$$1) \frac{x}{(x-1)(2x+3)}$$

$$11) \frac{1}{x^2 + a^2}$$

$$y = \frac{x}{(x-1)(2x+3)} = \frac{1/5}{x-1} + \frac{3/5}{2x+3} \quad (\because \text{partial fraction})$$

$$y_n = \frac{1}{5} \frac{(-1)^n n! 1^n}{(x-1)^{n+1}} + \frac{3}{5} \frac{(-1)^n n! 2^n}{(2x+3)^{n+1}}$$

$$= \frac{(-1)^n n!}{5} \left[\frac{1}{(x-1)^{n+1}} + \frac{3 \times 2^n}{(2x+3)^{n+1}} \right]$$

$$1) y = \frac{1}{x^2 + a^2} = \frac{1}{(x+ai)(x-ai)} \quad (\because \text{P.F.})$$

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$$y = \frac{1}{2ai} \left[\frac{1}{x-ai} - \frac{1}{x+ai} \right]$$

$$y_n = \frac{1}{2ai} \left\{ \frac{(-1)^n n!}{(x-ai)^{n+1}} - \frac{(-1)^n n!}{(x+ai)^{n+1}} \right\}$$

$$y_n = \frac{(-1)^n n!}{2ai} \left[\frac{1}{(x-ai)^{n+1}} - \frac{1}{(x+ai)^{n+1}} \right]$$

Practice problems:- Find n^{th} order derivative

1) $\log(4x^2-1)$ (use P.F.)

2) $\frac{x+2}{x+1} + \log \frac{x+2}{x+1}$

3) $\cos 2x \cos 3x$ 4) $\sin^3 x \cos^2 x$

5) $e^{5x} \cos x \cos 3x$ 6) $\frac{x}{x^2+a^2}$

7) If $y = \sin px + \cos px$, P.T. $y_n = p^n [1 + (-1)^n \sin 2px]^{\frac{1}{2}}$
Hence S.T. $y_8(\pi) = (1/2)^{3/2}$ when $p = \frac{1}{4}$

8) $\sinh 2x \sin 4x$ (Hint: $\sinh 2x = \frac{e^{2x} - e^{-2x}}{2}$)

Leibnitz's thm for the n^{th} order derivatives of the product of two fun.

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If u & v be two funs of x then the n^{th} order derivative of uv is

$$(uv)_n = u_n v + n C_1 u_{n-1} v_1 + n C_2 u_{n-2} v_2 + n C_3 u_{n-3} v_3 + \dots + n C_{n-2} u_2 v_{n-2} + \dots + n C_n u v_n$$

[OR]

let $y = uv$ then

$$y_n = u_n v + \frac{n}{1!} u_{n-1} v_1 + \frac{n(n-1)}{2!} u_{n-2} v_2 + \frac{n(n-1)(n-2)}{3!} u_{n-3} v_3 + \dots + 1 \cdot u v_n$$

e.g. ① $y = e^x (2x+3)^3$

$$y_n = e^x (2x+3)^3 + \frac{n}{1!} e^x (2x+3)^2 \cdot 2 + \frac{n(n-1)}{2!} e^x (2x+3) \cdot 2 \cdot 2 + \frac{n(n-1)(n-2)}{3!} e^x \cdot 2 \cdot 2 \cdot 2$$

$$y_n = e^x [(2x+3)^3 + 6n(2x+3)^2 + 12n(n-1)(2x+3) + 8n(n-1)(n-2)]$$

② If $y = (\sin^{-1} x)^2$ s.t. $(1-x^2) y_{n+2} - (2n+1)x y_{n+1} - n^2 y_n = 0$
Hence find $(y_n)_0$

$$\Rightarrow y = (\sin^{-1} x)^2 \quad \left| \begin{array}{l} y_1^2 (1-x^2) = 4(\sin^{-1} x)^2 \\ y_1^2 (1-x^2) = 4y \\ \text{diff. w.r.t } x \end{array} \right.$$

$$(1-x^2)2y_1 y_2 + y_1^2(-2x) = 4y_1$$

divided by $2y_1$

$$(1-x^2)y_2 - 2xy_1 = 2$$

Diff. n^{th} times w.r.t. x

$$\left[y_{n+2}(1-x^2) + n y_{n+1}(-2x) + \frac{n(n-1)}{2!} y_n(-2) \right] - 2 \left[y_{n+1}x + n y_n \cdot 1 \right] = 0$$

$$(1-x^2)y_{n+2} - 2xny_{n+1} - n^2y_n + ny_n - 2xy_{n+1} - 2ny_n = 0$$

$$(1-x^2)y_{n+2} - 2nx y_{n+1} - 2xy_{n+1} - n^2y_n + ny_n - 2ny_n = 0$$

$$(1-x^2)y_{n+2} - 2xy_{n+1}(n+1) - n^2y_n = 0$$

$$\boxed{(1-x^2)y_{n+2} - x(2n+1)y_{n+1} - n^2y_n = 0} \quad \text{--- (1)}$$

put $x=0$

$$(y_{n+2})_0 = n^2(y_n)_0$$

$$(y_1)_0 = 0, (y_2)_0 = 2, (y_3)_0 = 1^2(y_1)_0 = 0$$

$$(y_4)_0 = 2^2(y_2)_0 = 2^2 \cdot 2$$

$$(y_5)_0 = 3^2(y_3)_0 = 0$$

$$(y_6)_0 = 4^2(y_4)_0 = 4^2 \cdot 2^2 \cdot 2$$

$$(y_n)_0 = \begin{cases} 0, & n = \text{odd} \\ (n-2)^2(n-4)^2(n-6)^2 \dots 2, & n = \text{even} \end{cases}$$

③ If $y = e^{a \sin^{-1} x}$ P.T. $(1-x^2)y_{n+2} - (2n+1)x y_{n+1} - (n^2+a^2)y_n = 0$
Find $(y_n)_0$

$$\Rightarrow y = e^{a \sin^{-1} x}$$

$$y_1 = \frac{a e^{\sin^{-1} x}}{\sqrt{1-x^2}}$$

$$\sqrt{1-x^2} y_1 = a y$$

$$(1-x^2) y_1^2 = a^2 y^2$$

diff. w.r.t. x

$$(1-x^2) 2 y_1 y_2 - 2x y_1^2 = 2a^2 y y_1$$

$$(1-x^2) y_2 - x y_1 = a^2 y$$

Diff. n times w.r.t. x by L.T. we get

$$(1-x^2) y_{n+2} - (2n+1)x y_{n+1} - (n^2+a^2) y_n = 0$$

$$\text{also } (y_n)_0 = a^2 (2^2+a^2) (4^2+a^2) \dots [(n-2)^2+a^2]$$

④ If $y^{1/m} + \bar{y}^{1/m} = 2x$ P.T. $(x^2-1)y_{n+2} + (2n+1)x y_{n+1} + (n^2-m^2)y_n = 0$

$$\Rightarrow y^{1/m} + \bar{y}^{1/m} = 2x$$

$$(y^{1/m})^2 - 2x y^{1/m} + 1 = 0$$

$$y^{1/m} = \frac{-(-2x) \pm \sqrt{4x^2-4}}{2} = x \pm \sqrt{x^2-1}$$

$$\therefore y = [x \pm \sqrt{x^2-1}]^m$$

$$\log y = m \log [x \pm \sqrt{x^2-1}]$$

diff w.r.t. x

$$\frac{1}{y} y_1 = m \cdot \frac{1 \pm \frac{x}{\sqrt{x^2-1}}}{x \pm \sqrt{x^2-1}} \left[1 \pm \frac{x}{\sqrt{x^2-1}} \right]$$

$$\frac{y_1}{y} = \frac{m}{x \pm \sqrt{x^2 - 1}} \left[\frac{x \pm \sqrt{x^2 - 1}}{\sqrt{x^2 - 1}} \right]$$

$$\sqrt{x^2 - 1} y_1 = m y$$

$$(x^2 - 1) y_1^2 = m^2 y^2$$

diff. w.r.t. x

$$(x^2 - 1) 2 y_1 y_2 + 2 x y_1^2 = m^2 \cdot 2 y y_1$$

$$(x^2 - 1) y_2 + x y_1 = m^2 y$$

diff. n times w.r.t. x using L.R. we get

$$(x^2 - 1) y_{n+2} + (2n+1) x y_{n+1} + (n^2 - m^2) y_n = 0$$

Practice problems : [Book: B.S. Grewal P.N. 141]
problems: 4.3

1) If $y = a \cos(\omega g x) + b \sin(\omega g x)$, S.T.

$$x^2 y_2 + x y_1 + y = 0 \quad \& \quad x^2 y_{n+2} + (2n+1) x y_{n+1} + (n^2 + 1) y_n = 0$$

2) If $y = \sin^{-1} x$ P.T. $(1 - x^2) y_{n+2} - (2n+1) x y_{n+1} - n^2 y_n = 0$
also find $(y_n)_0$

3) If $\cos^{-1}(y/b) = \omega g(\frac{x}{n})$, P.T. $x^2 y_{n+2} + (2n+1) x y_{n+1} + 2n^2 y_n = 0$

4) If $y = \tan^{-1} x$ P.T. $(1 + x^2) y_{n+1} + 2n x y_n + n(n-1) y_{n-1} = 0$
also find $(y_n)_0$

5) If $y = \cos(m \sin^{-1} x)$ P.T. $(1 - x^2) y_{n+2} - (2n+1) x y_{n+1} + (m^2 - n^2) y_n = 0$

6) If $y = \sin(m \sin^{-1} x)$ P.T. $(1 - x^2) y_{n+2} - 2(n+1) x y_{n+1} + (m^2 - n^2) y_n = 0$

7) If $y = e^{m \cos^{-1} x}$ P.T. $(1 - x^2) y_{n+2} - (2n+1) x y_{n+1} - (n^2 + m^2) y_n = 0$

$$8) y = (x^2-1)^n \text{ P.T. } (x^2-1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0$$

$$9) \text{ If } \sin y = 2 \cos y (x+1) \text{ P.T. } (x+1)^2 y_{n+2} + (2n+1)(x+1)y_{n+1} + (x^2+4)y_n = 0$$

$$10) \text{ If } y = x^n \cos x \text{ P.T. } y_{n+1} = \frac{n!}{x}$$

$$11) v = x^n \cos x \text{ S.T. } (v_n = n v_{n-1} + (n-1)!) \Rightarrow \text{verdone}$$

$$v_1 = x^n \cdot \frac{1}{x} + n x^{n-1} \cos x = x^{n-1} + \frac{n}{x} x^n \cos x$$

$$v_1 = x^{n-1} (1 + n \cos x) \quad | \quad v_1 = x^{n-1} + \frac{n}{x} v$$

$$v_1 = \frac{x^n}{x} + \frac{n}{x} v$$

$$v_1 x = x^n + n v$$

diff. n times w.r.t. x

$$v_{n+1} x + n v_n \cdot 1 = n! + n v_n$$

$$v_{n+1} = \frac{n!}{x}$$

$$v_{n+1} = \frac{(n-1)!}{x}$$

$$v_{n-1} = \frac{(n-2)!}{x}$$

$$y v_n =$$

$$v_n = n! \cos x + (n-1)! \cdot \frac{1}{x} + \frac{n(n-1)(n-2)!}{2!}$$

$$12) y = \sin y (x^2 + 2x + 1) \text{ P.T. } (x+1)^2 y_{n+2} + (2n+1)(x+1)y_{n+1} + (n^2+4)y_n = 0$$

$$13) x = \sin t, y = \cos pt \text{ S.T. } (1-x^2)y_{n+2} - (2n+1)x y_{n+1} - (n^2-p^2)y_n = 0$$

$$14) y = \cos(m \cos x) \text{ P.T. } x^2 y_{n+2} + (2n+1)x y_{n+1} + (m^2+n^2)y_n = 0$$

Taylor's Theorem :-

If i) $f(x)$ & its first $(n-1)$ derivatives be continuous in $[a, a+h]$ & ii) $f^n(x)$ exists for every value of x in $(a, a+h)$, then there is at least one no. θ ($0 < \theta < 1$) s.t.

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^n}{n!} f^n(a + \theta h)$$

if $\theta = 1$ or $\theta = 0$ & $x = a$ or $x = a+h$

i) if $a = x$

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \dots + \frac{h^n}{n!} f^n(x) + \dots$$

ii) if $h = (x-a)$ & $x = a$

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots$$

iii) if $h = x$ & $a = 0$

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

which is called as Maclaurin's series

Maclaurin's series :- If $f(x)$ can be expanded as an infinite series then

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

① Expand $\tan x$ using M.S. upto containing the term x^5

$$\Rightarrow f(x) = \tan x$$

$$f'(x) = \sec^2 x = 1 + \tan^2 x = 1 + [f(x)]^2$$

$$f''(x) = 2 f(x) f'(x)$$

$$f'''(x) = 2 [f(x) f''(x) + [f'(x)]^2]$$

$$f^{IV}(x) = 2 \{ f(x) f'''(x) + f''(x) f'(x) + 2 f'(x) f''(x) \}$$

$$f^V(x) = 2 \{ f(x) f^{IV}(x) + f'''(x) f'(x) + 3 [f''(x)]^2 + f'(x) f'''(x) \}$$

$$f(0) = 0$$

$$f'(0) = 1 + 0 = 1$$

$$f''(0) = 0$$

$$f'''(0) = 2 [0 + 1^2] = 2$$

$$f^{IV}(0) = 2 \{ 0 + 0 + 0 \} = 0$$

$$f^V(0) = 2 \{ 0 + 1 \times 2 + 3 [0 + 1 \times 2] \} = 2 \{ 2 + 6 \} = 16$$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{IV}(0) + \frac{x^5}{5!} f^V(0) + \dots$$

$$f(x) = 0 + x \times 1 + 0 + \frac{x^3}{6} \times 2 + 0 + \frac{x^5}{120} \times 16 + \dots$$

$$f(x) = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$$

② Expand $e^{\sin x}$ by M.S. upto the term containing x^4
Ans: $1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \dots$

③ Expand $\log(1 + \sin^2 x)$ upto x^6

As we know that, the M.S. of $\log(1+t)$

$$\log(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \dots$$

$$\therefore \sin^2 x = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

$$\therefore \sin^2 x = \left[x - \left(\frac{x^3}{6} - \frac{x^5}{120} + \dots \right) \right]^2$$

$$\downarrow$$

$$+ = x^2 - 2x \left(\frac{x^3}{6} - \frac{x^5}{120} + \dots \right) + \left(\frac{x^3}{6} - \frac{x^5}{120} + \dots \right)^2$$

$$= x^2 - \frac{x^4}{3} + \frac{x^6}{60} + \dots + \frac{x^6}{36} + \dots$$

$$+ = x^2 - \frac{x^4}{3} + \frac{2x^6}{45} \quad \text{use in series of } \log(1+t)$$

$$\therefore \log(1 + \sin^2 x) = \left(x^2 - \frac{x^4}{3} + \frac{2x^6}{36} + \dots \right) - \frac{1}{2} \left(x^2 - \frac{x^4}{3} + \frac{2x^6}{36} + \dots \right)^2 + \frac{1}{3} \left(x^2 - \frac{x^4}{3} + \frac{2x^6}{36} + \dots \right)^3 - \dots$$

$$= x^2 - \frac{x^4}{3} + \frac{2x^6}{36} - \frac{x^4}{2} + \frac{x^6}{3} + \dots$$

$$- \frac{1}{2} \left(x^4 - \frac{2x^6}{3} + \dots \right) + \frac{x^6}{3} + \dots$$

$$\log(1 + \sin^2 x) = x^2 - \frac{5}{6} x^4 + \frac{32}{45} x^6 + \dots$$

3 (4) Expand $e^{a \sin^2 x}$ with M.S.

$$\text{Ans: } 1 + ax + \frac{a^2 x^2}{2!} + \frac{a(1+a^2)}{3!} x^3 + \frac{a^2(2+a^2)}{4!} x^4 + \dots$$

(5) Expand $\log x$ in power of $(x-1)$ & hence evaluate $\log 1.1$ correct to 4 decimal place.

$$\Rightarrow f(x) = f(1) + \frac{(x-1)}{1!} f'(1) + \frac{(x-1)^2}{2!} f''(1) + \dots \quad \text{--- (1)}$$

$$f(x) = \log x$$

$$f'(x) = \frac{1}{x}$$

$$f''(x) = -\frac{1}{x^2}$$

$$f(1) = 0$$

$$f'(1) = 1$$

$$f''(1) = -1$$

$$f'''(x) = 2 \cdot \frac{2}{x^3}$$

$$f^{IV}(x) = -\frac{6}{x^4}$$

$$f'''(1) = 2$$

$$f^{IV}(1) = -6$$

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$$f(x) = 0 + (x-1)(1) + \frac{(x-1)^2}{2!}(-1) + \frac{(x-1)^3}{3!}(2) + \frac{(x-1)^4}{4!}(-6) + \dots$$

$$\log x = (x-1) - \frac{(x-1)^2}{2} + \frac{1}{3}(x-1)^3 - \frac{(x-1)^4}{4} + \dots$$

put $x = 1.1$ $\therefore x-1 = 1.1 - 1 = 0.1$

$$\log 1.1 = (0.1) - \frac{(0.1)^2}{2} + \frac{(0.1)^3}{3} - \frac{(0.1)^4}{4} + \dots$$

$$\boxed{\log 1.1 = 0.0953}$$

Practice Problems : $\Rightarrow$

* Using Maclaurin's series, expand the foll. funs.

- 1) $\log(1+x)$ 2) $\sin^2 x$ 3) $\tan^2 x$ 4) $\log \sec x$
- 5) $e^x \sin x$ 6) $\sec x$ 7) $e^{\cos^2 x}$ 8) $e^{x \sin x}$
- 9) $\log(1+\sin x)$ 10) $\log(1+e^x)$ 11) $\sin x$ 12) $\cos x$
- 13) e^x 14) $\log(1-x)$ 15) $\log \sec x$

* By forming D.E. (w/o L.R.) S.T.

$$1) (\sin^2 x)^2 = 2 \frac{x^2}{2!} + 2 \cdot 2^2 \cdot \frac{x^4}{4!} + 2 \cdot 2^2 \cdot 4^2 \cdot \frac{x^6}{6!} + \dots$$

$$2) y = \sin(m \sin^2 x) \text{ S.T. } (1-x^2)y_2 - xy_1 + m^2y = 0$$

Hence expand $\sin m \sin^2 x$ in power of $\sin x$

3) 1) Using Taylor form express $2x^3 + 7x^2 + x - 6$ in power of $(x-1)$

2) Expand e^x & $\tan^2 x$ in power of $(x-1)$ using T.S.

3) Expand $\sin x$ in power of $(x - \pi/2)$

4) P-T. $\cos x = \cos a + (x-a) \cos a - \frac{1}{2}(x-a)^2 \cos a + \dots$

5) Find T.S. expansion for $i) \frac{1}{x}$ about 1

ii) $\cos x$ about $\pi - \pi/3$ i.e. in power of $(x - \pi/3)$

6) compute to four decimal places the value of $\cos 32^\circ$ by T.S.

~~7) Find T.S. expansion for $\cos x$ about $\pi/3$~~

Indeterminate Form's

$$\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \times \infty, 0^0, 1^\infty, \infty^0$$

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \quad \text{if } f(a) = 0 \text{ \& } g(a) = 0$$

i.e. diff. N^r & D^r separately

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} (1+x)^{1/x} = e$$

1) Evaluate $\lim_{x \rightarrow 0} \frac{x e^x - \cos(1+x)}{x^2}$

$$L = \lim_{x \rightarrow 0} \frac{x e^x - \cos(1+x)}{x^2} \quad \left(\frac{0}{0} \text{ form} \right)$$

$\equiv$ use L-H-R

$$L = \lim_{x \rightarrow 0} \frac{x e^x + e^x - \frac{1}{1+x}}{2x} \quad \left(\frac{0}{0} \text{ form} \right)$$

use L-H-R

$$= \lim_{x \rightarrow 0} \frac{x e^x + e^x + e^x + 1/(1+x)^2}{2}$$

$$L = \frac{0 + 1 + 1 + 1}{2} = \frac{3}{2}$$

$$2) L = \lim_{x \rightarrow 0} \frac{x^x - x}{x - 1 - \log x} \quad \left(\frac{0}{0} \text{ form} \right)$$

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Use L-H-R

$$L = \frac{x^x (1 + \log x) - 1}{1 - \frac{1}{x}} \quad \left(\frac{0}{0} \text{ form} \right)$$

$$\text{Let } y = x^x$$

$$\log y = x \log x$$

$$\frac{1}{y} y' = 1 + \log x$$

$$y' = x^x (1 + \log x)$$

$$L = \frac{x^x \cdot \frac{1}{x} + x^x (1 + \log x)^2}{+1/x^2}$$

$$L = \frac{1 + 1}{+1} = +2$$

$$\boxed{L = 2}$$

3) Find the value of a, b, c & t.

$$\lim_{x \rightarrow 0} \frac{x(a + b \cos x) - c \sin x}{x^3} = 1$$

$$\text{Soln: } \lim_{x \rightarrow 0} \frac{x(a + b \cos x) - c \sin x}{x^3} = \left(\frac{0}{0} \text{ form} \right)$$

using L-H-R

$$L = \lim_{x \rightarrow 0} \frac{x(-b \sin x) + (a + b \cos x) - c \cos x}{3x^2} \quad \text{--- (1)}$$

As $0^0 \rightarrow 0$ for $x=0$, can (1) $\rightarrow$ finite limit iff

$$N^0 \rightarrow 0 \text{ for } x=0$$

$$\therefore a + b - c = 0 \quad \text{--- (2)}$$

using L-H-R

$$L = \frac{-b \sin x - b x \cos x + -b \sin x + c \sin x}{20x^3}$$

$$L = \frac{-2b \sin x - b x \cos x + c \sin x}{20x^3}$$

$$L = \frac{\sin x (c - 2b) - b x \cos x}{20x^3}$$

$$\left(\frac{0}{0} \right)$$

using LHR

$$L = \lim_{n \rightarrow 0} \frac{(c-2b) \sin n - b(\cos n - n \sin n)}{60n^2}$$

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As $n \rightarrow 0$ $\sin n \rightarrow 0$ $\therefore N^r \rightarrow 0$ as $n \rightarrow 0$

$$\therefore c-2b-b=0$$

$$c-3b=0$$

$$\therefore \boxed{c=3b} \quad \text{--- (3)}$$

using LHR

$$L = \lim_{n \rightarrow 0} \frac{-(c-2b) \sin n - b[-\sin n - \sin n - n \cos n]}{120n}$$

$$= \lim_{n \rightarrow 0} \frac{(2b-c) \sin n + 2b \sin n + b n \cos n}{120n}$$

$$= \lim_{n \rightarrow 0} \frac{(4b-c) \sin n + b n \cos n}{120n} \quad \left(\frac{0}{0} \text{ f}\right)$$

using LHR

$$L = \lim_{n \rightarrow 0} \frac{(4b-c) \cos n + b[\cos n - n \sin n]}{120}$$

but $L=1$ as $n \rightarrow 0$ & $0^r = 120$ $\therefore N^r = 120$ as $n \rightarrow 0$

$$4b-c+b[1-0]=120$$

$$5b-c=120$$

$$\Rightarrow 5b-3b=120 \quad (\because (3))$$

$$2b=120$$

$$\boxed{b=60}$$

$$\therefore \boxed{c=3 \times 60 = 180}$$

$$(2) \Rightarrow a+b-c=0$$

$$a+60-180=0$$

$$\boxed{a=120}$$

$$a=120$$

$$b=60$$

$$c=180$$

4) Evaluate $\lim_{x \rightarrow 0} \frac{\log x}{\cot x}$

$$L = \lim_{x \rightarrow 0} \frac{\log x}{\cot x} \quad \left(\frac{0}{0} \text{ f} \right)$$

use LHR

$$L = \lim_{x \rightarrow 0} \frac{1/x}{-\operatorname{cosec}^2 x} = - \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2} \cdot x = 0$$

5) $\lim_{x \rightarrow 0} L = \lim_{x \rightarrow 0} \frac{e^x \sin x - x - x^2}{x^2 + x \log(1-x)} \quad \left(\frac{0}{0} \text{ f} \right)$

LHR

$$L = \lim_{x \rightarrow 0} \frac{e^x \cos x + e^x \sin x - 1 - 2x}{2x + x(-1) + \log(1-x)}$$

$$L = \lim_{x \rightarrow 0} \frac{e^x (\cos x + \sin x) - 1 - 2x}{2x - 2x - 2x^2 - x + \log(1-x)}$$

$$L = \lim_{x \rightarrow 0} \frac{e^x (\cos x + \sin x) - 1 - 2x}{\frac{2x^2 - x}{x-1} + \log(1-x)} \quad \left(\frac{0}{0} \text{ f} \right)$$

LHR

$$L = \lim_{x \rightarrow 0} \frac{e^x (\cos x - \sin x) + e^x (\cos x + \sin x) - 2}{(x-1)(4x-1) - (2x^2 - x) \cdot 1} = \frac{1}{(x-1)^2}$$

$$L = \frac{e^x (\cos x - \sin x) + e^x (\cos x + \sin x) - 2}{4x^2 - 4x + 1 - 2x^2 + x} = \frac{2e^x \cos x - 2}{(x-1)^2}$$

$$L = \frac{2e^x \cos x - 2}{(x-1)^2} = \frac{(2e^x \cos x - 2)(x-1)^2}{(x-1)^2} = \frac{2x^2 - 4x + 1}{(x-1)^2}$$

$$L = \frac{(2e^x \cos x - 2)(x-1)^2}{(x^2 - 2x) \cdot x^2 - 2x}$$

$$L = \lim_{n \rightarrow 0} \frac{(e^n \cos n - 1)(n-1)^2}{n^2 - 2n} \quad \left(\frac{0}{0} \text{ f}\right)$$

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LHR

$$L = \lim_{n \rightarrow 0} \frac{(e^n \cos n - 1) \cdot 2(n-1) + (n-1)^2 [e^n \cos n - e^n \sin n]}{2n - 2}$$

$$L = \frac{0 + (-1)^2 [1 - 0]}{-2} = -\frac{1}{2}$$

(OR)

$$L = \lim_{n \rightarrow 0} \frac{e^n \sin n - n - n^2}{n^2 + n \log(1-n)}$$

$$= \lim_{n \rightarrow 0} \frac{(1 + n + \frac{n^2}{2!} + \frac{n^3}{3!} + \dots)(n - \frac{n^3}{3!} + \dots) - n - n^2}{n^2 + n[-n - \frac{n^2}{2} - \frac{n^3}{3} - \dots]}$$

$$= \lim_{n \rightarrow 0} \frac{[n + n^2 + \frac{n^3}{2!} + \frac{n^4}{3!} + \dots - \frac{n^3}{3!} - \frac{n^4}{3!} + \dots] - n - n^2}{n^2 - n^2 - \frac{n^3}{2} - \frac{n^4}{3} - \dots}$$

$$= \lim_{n \rightarrow 0} \frac{\frac{n^3}{2!} - \frac{n^3}{3!} + \frac{n^4}{3!} - \frac{n^4}{3!} + \dots}{-\frac{n^3}{2} - \frac{n^4}{3} - \dots}$$

$$= \lim_{n \rightarrow 0} \frac{\frac{n^3}{3} + \dots}{-\frac{n^3}{2} - \frac{n^4}{3} - \dots} = \frac{n^3 \left[\frac{1}{3} + \dots \right]}{n^3 \left[-\frac{1}{2} - \frac{1}{3}n - \dots \right]}$$

$$= \frac{1/3}{-1/2}$$

$$= -\frac{2}{3}$$

$$\star \lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x}$$

$$y = (1+x)^{1/x}$$

$$\log y = \frac{1}{x} \log(1+x)$$

$$= \frac{1}{x} \left[x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \right]$$

$$\log y = 1 - \frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots$$

$$y = e^{1 - \frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots} = e \cdot e^{-\frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots}$$

$$\therefore L = \lim_{x \rightarrow 0} \frac{e \cdot e^{-\frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots} - e}{x} \quad \left(\frac{0}{0} \text{ f} \right)$$

LHR

$$L = \lim_{x \rightarrow 0} \frac{e \cdot e^{-\frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots} \cdot \left(-\frac{1}{2} + \frac{2x}{3} - \dots \right) - 0}{1}$$

$$L = \frac{e \cdot e^0 \left(-\frac{1}{2} + 0 - \dots \right)}{1} = \frac{-e/2}{1} = -\frac{e}{2}$$

Practice problems: (P.N. 154 B.S. Grewal)

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{a^x - b^x}{x} \quad \textcircled{2} \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x^2 \sin x} \quad \textcircled{3} \lim_{x \rightarrow 0} \frac{x - \sin x}{\sin x (1 - \cos x)}$$

$$\textcircled{4} \lim_{x \rightarrow \pi/2} \frac{a^{\sin x} - a}{\log \sin x} \quad \textcircled{5} \lim_{x \rightarrow 0} \frac{\sin x \sin^{-1} x - x^2}{x^6} \quad \textcircled{6} \lim_{x \rightarrow 0} \frac{\log \sec x - x^2/2}{x^4}$$

$$\textcircled{7} \lim_{x \rightarrow 0} \frac{x \cos x - \log(1+x)}{x^2} \quad \textcircled{8} \lim_{x \rightarrow 0} \frac{x e^x - \log(1+x)}{x^2} \quad \textcircled{9} \lim_{x \rightarrow a} \frac{\log(x-a)}{\log(e^x - e^a)}$$

$$\textcircled{9} \lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e - \frac{1}{2} e x}{x^2} \quad \textcircled{10} \text{ Find } a, b \text{ if } \lim_{x \rightarrow 0} \frac{\sin 2x + a \sin x}{x^2} \text{ is finite}$$

$$\textcircled{11} \text{ Find } a, b \text{ if } \lim_{x \rightarrow 0} \frac{a \sinh x + b \sin x}{x^3} = \frac{5}{3} \quad \textcircled{14} \lim_{x \rightarrow 0} \frac{\tan x - x}{x^2 + 4 \sin x}$$

$$\textcircled{12} \text{ Find } a, b, c \text{ if } \lim_{x \rightarrow 0} \frac{a e^x - b \cos x + c e^{-x}}{x \sin x} = 2$$

Other forms Reducible to $\frac{0}{0}$ or $\frac{\infty}{\infty}$

$$1) \infty - \infty = \frac{1}{0} - \frac{1}{0} = \frac{0-0}{0} = \frac{0}{0}$$

$$2) 0 \times \infty = 0 \times \frac{1}{0} = \frac{0}{0}$$

$$3) \text{ if } L = 0^\circ \Rightarrow \log L = 0 \log 0 = 0 \times \infty = \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

$$4) L = 1^\circ \Rightarrow \log L = \infty \log 1 = \infty \times 0 = \frac{1}{0} \times 0 = \frac{0}{0}$$

$$5) L = \infty^\circ \Rightarrow \log L = 0 \log \infty = 0 \times \infty = \frac{1}{\infty} \times \infty = \frac{\infty}{\infty}$$

$$1) \lim_{x \rightarrow 0} \frac{1}{\sin x} - \frac{1}{x} \neq$$

$$L = \lim_{x \rightarrow 0} \frac{x - \sin x}{x \sin x} = \lim_{x \rightarrow 0} \frac{x - \sin x}{x^2} \left(\frac{0}{0} \right) \left(\because \sin x = 1 \right)$$

$$= \frac{1 - \cos x}{2x} \left(\frac{0}{0} \right)$$

$$= \frac{\sin x}{2}$$

$$L = 0$$

$$2) \lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x} \quad (1^\infty)$$

$$L = \lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x}$$

$$\log L = \lim_{x \rightarrow \frac{\pi}{2}} \tan x \cdot \log \sin x = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\log \sin x}{\cot x} \quad \left(\frac{0}{0} \right)$$

LHR

$$\log L = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cot x}{-\csc^2 x} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x / \sin x}{-1 / \sin^2 x} = -\sin x \cos x$$

$$\log L = 0 \quad \therefore L = e^0 = 1$$

$$\textcircled{3} \lim_{n \rightarrow \infty} \left(\frac{a^n + b^n + c^n}{3} \right)^{1/n} \quad (1^\infty)$$

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$$\text{Wgt} = \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{a^n + b^n + c^n}{3} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{\log(a^n + b^n + c^n) - \log 3}{n} \quad \left(\frac{0}{0} \right)$$

$$\stackrel{\text{L'H}}{=} \lim_{n \rightarrow \infty} \frac{a^n \log a + b^n \log b + c^n \log c}{a^n + b^n + c^n} - 0$$

$$= \frac{\log a + \log b + \log c}{3}$$

$$= \frac{1}{3} \log abc$$

$$\text{Wgt} = \log(abc)^{1/3}$$

$$L = e^{\log(abc)^{1/3}}$$

$$L = (abc)^{1/3}$$

$$\textcircled{4} L = \lim_{n \rightarrow \infty} \left(\frac{\tan n}{n} \right)^{1/n^2}$$

$$\text{Wgt} L = \lim_{n \rightarrow \infty} \frac{1}{n^2} \log \left(\frac{\tan n}{n} \right) \quad \left(\frac{0}{0} \right) \quad \left(\frac{0}{0} \right)$$

$$= \frac{n}{\tan n} \lim_{n \rightarrow \infty} \log \left(\frac{\sin n}{n} \cdot \frac{1}{\cos n} \right) = \frac{\log \sec n}{n^2} \cdot \frac{1}{\sin n} \quad \left(\because \frac{\sin n}{n} = 1 \right)$$

$$= \frac{1}{\sec n} \cdot \frac{\sec n \tan n}{n^2} = \frac{\tan n}{n^2} \quad \left(\frac{0}{0} \right)$$

$$= \frac{\sec^2 n}{2n}$$

$$\stackrel{\text{L'H}}{=} \lim_{n \rightarrow \infty} \frac{\frac{n}{\tan n} \left(\frac{n \sec^2 n - \tan n}{n^2} \right)}{2n} = \frac{1 \times (n \sec^2 n - \tan n)}{2n^3}$$

$$\left(\because \frac{n}{\tan n} = 1 \text{ as } n \rightarrow \infty \right)$$

$$\log L = \frac{x \sec^2 x - \tan x}{2x^3} \text{ as } x \rightarrow 0 \left(\frac{0}{0} \right)$$

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$$yL\& = \frac{x \cdot 2 \sec^2 x \cdot \tan x + \sec^2 x - \sec^2 x}{6x^2}$$

$$\log L = \frac{2x \sec^2 x \tan x}{6x^2}$$

$$\therefore \log L = \lim_{x \rightarrow 0} \frac{2}{3} \sec^2 x \left(\frac{\tan x}{x} \right)$$

$$= \frac{1}{3} \sec^2 0 \times 1$$

$$\log L = \frac{1}{3}$$

$$L = e^{1/3}$$

Practice Problems, BS Grewal P.N. = 156

① $\lim_{x \rightarrow 0} \frac{1}{x^2} - \frac{1}{\sin^2 x}$

② $\lim_{x \rightarrow 0} \frac{1}{x} - \frac{1}{e^x - 1}$

③ $\lim_{x \rightarrow 1} 2x \tan x - \pi \sec x$

④ $\lim_{x \rightarrow 0} \frac{\cot x - \frac{1}{x}}{x}$

⑤ $\lim_{x \rightarrow 0} \frac{1}{x^2} - \cot^2 x$

⑥ $\lim_{x \rightarrow 1} x \frac{1}{1-x}$

⑦ $\lim_{x \rightarrow 0} (a^x + x)^{1/x}$

⑧ $\lim_{x \rightarrow \pi/2} (\sec x)^{\cot x}$

⑨ $\lim_{x \rightarrow 0} (1 + \sin x)^{\cot x}$

⑩ $\lim_{x \rightarrow 0} (\cos x)^{1/x^2}$

⑪ $\lim_{x \rightarrow \pi/2} (\tan x)^{\tan 2x}$

⑫ $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x}$

⑬ $\lim_{x \rightarrow \pi/2} (\cos x)^{\frac{\pi}{2} - x}$

⑭ $\lim_{x \rightarrow 0} \frac{\sin x}{x} (1-x^2)^{\frac{1}{\log(1-x)}}$

⑮ $\lim_{n \rightarrow \infty} \left(\frac{1^n + 2^n + 3^n + 4^n}{3} \right)^{1/n}$

$$\log L = 0 \quad \therefore L = e^0 = 1$$